КР 1.

Вариант 1.

1. $X \sim U[0;10]$

$W = 10$

$U(w) = \sqrt{w}$

$U(W - G) = E[U(W - X)]$

$G - ?$

$$E[U(W-X)] = \int_0^{10} \sqrt{10-x} \cdot \frac{1}{10}\,dx =$$

$$= \frac{1}{10}\left(-\frac{2}{3}(10-x)^{\frac{3}{2}}\right)\Bigg|_0^{10} = \frac{1}{10}\left(\frac{2}{3}\,10^{\frac{3}{2}}\right) =$$

$$= \frac{2}{3}\sqrt{10}$$

$$\sqrt{10-G} = \frac{2}{3}\sqrt{10}$$

$$10-G = \frac{4}{9}\cdot 10$$

$$G = 10\cdot\frac{5}{9} = \frac{50}{9} = 5,\overline{5}$$

2 $p=0,03$ $\quad f(x) = 3x^2 \cdot I_{[0;1]}(x)$

$E[X]$, $D[X]$ -?

$P\{\sum^{100} X_i > 1,05\, E[X]\}$ -?

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a) $X = I\cdot B$

$$y = \sum^{100} X_i$$

$$E[B] = \int_0^1 x \cdot 3x^2\,dx = 3\left(\frac{x^4}{4}\right)\Big|_0^1 = \frac{3}{4} = \mu$$

$$E[B^2] = \int_0^1 x^2\, 3x^2\,dx = 3\left(\frac{x^5}{5}\right)\Big|_0^1 = \frac{3}{5}$$

$$D[B] = \frac{3}{5} - \frac{9}{16} = \frac{3}{80} \approx 0{,}0375 = \sigma^2$$

$$E[X] = p \cdot \mu = \frac{9}{400}$$

$$D[X] = D[I \cdot B] = D[E[X|I]] + E[D[X|I]]$$

$$E[X|I] = \mu I$$

$$D[\mu I] = \mu^2 \cdot p(1-p)$$

$$D[X|I] = \sigma^2 I$$

$$E[\sigma^2 I] = \sigma^2 p$$

$$\Rightarrow D[X] = \mu^2 p(1-p) + \sigma^2 p =$$

$$= \left(\frac{3}{4}\right)^2 \cdot \frac{3}{100} \cdot \frac{97}{100} + \frac{3}{80} \cdot \frac{3}{100} =$$

$$= \frac{2799}{160\,000} \approx \boxed{0{,}0174937}$$

$$Y = \sum^{100} X_i,$$

$$E[Y] = 100\, E[X] = 100 \cdot \frac{9}{400} = \frac{9}{4}$$

$$D[Y] = 100\, D[X] = 100 \cdot \ldots = 1{,}74937$$

б) $Y \sim N\left(\frac{9}{4}, 1{,}74937\right)$

$$P\{Y > 1{,}05\, E[Y]\} =$$

$$= P\left\{ \frac{Y - E[Y]}{\sigma[Y]} > \frac{0{,}05\, E[Y]}{\sigma[Y]} \right\} =$$

$$= 1 - \Phi\left(\frac{0{,}05\, E[Y]}{\sigma[Y]} \right) =$$

$$= 1 - \Phi\left(\frac{0{,}05 \cdot 9/4}{\sqrt{1{,}74937}} \right) =$$

$$= 1 - \Phi(0{,}085)$$

3. $X_1 \sim E(3)$, $X_2 \sim E(1)$

$$f_S(x) = \int_0^x 3e^{-3(x-t)} e^{-t}\, dt =$$

$$= 3\int_0^x e^{-3x+3t-t}\, dt = 3e^{-3x}\int_0^x e^{2t}\, dt =$$

$$= 3e^{-3x}\left(\frac{1}{2}e^{2t}\right)\Big|_0^x = 3e^{-3x}\left(\frac{1}{2}e^{2x} - \frac{1}{2}\right)$$

$$= \frac{3}{2}e^{-x} - \frac{3e^{-3x}}{2} = \frac{3}{2}e^{-3x}\left(e^{2x} - 1\right)$$

Вариант 2

1) $X \sim E(3)$, $w=30$, $U(w) = -e^{-0,5w}$

$$E[U(w-X)] = \int_0^{+\infty} -e^{-0,5(w-X)} \cdot 3e^{-3X} dx =$$

$$= -3 \int_0^{+\infty} e^{-0,5w+0,5X-3X} dx =$$

$$= -3e^{-0,5w} \int_0^{+\infty} e^{-2,5X} dx = -3e^{-0,5w} \left(-\frac{1}{2,5} e^{-2,5X} \right) \Big|_0^{+\infty} =$$

$$= -3e^{-0,5w} \left(0 + \frac{1}{2,5} \right) = -\frac{6}{5} e^{-0,5w}$$

$$U(w-G) = -e^{-0,5(w-G)} = -e^{-0,5w} \cdot e^{0,5G}$$

$$-e^{-0,5w} \cdot e^{0,5G} = -\frac{6}{5} e^{-0,5w}$$

$$e^{0,5G} = \frac{6}{5}$$

$$0,5G = \ln \frac{6}{5}$$

$$G = 2 \ln 6/5$$

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② $p = 0,01$ $\quad f(x) = 5x^4 \, I_{[0,1]}(x)$

$X = IB$

$$E[B] = \int_0^1 x \cdot 5x^4 \, dx = 5\left(\frac{x^6}{6}\right)\Big|_0^1 = \frac{5}{6}$$

$$E[B^2] = \int_0^1 x^2 \cdot 5x^4 \, dx = 5\left(\frac{x^7}{7}\right)\Big|_0^1 = \frac{5}{7}$$

$$D[B] = \frac{5}{7} - \left(\frac{5}{6}\right)^2 = \frac{5}{252} \approx 0,01984$$

$$D[X] = \mu^2 p(1-p) + \sigma^2 p =$$

$$= \left(\frac{5}{6}\right)^2 \cdot \frac{1}{100} \frac{99}{100} + \frac{5}{252} \cdot \frac{1}{100} \approx 0{,}00707$$

$$E[X] = \frac{1}{100} \cdot \frac{5}{6} = \frac{1}{120}$$

$$Y = \sum^{50} X_i$$

$$E[Y] = \frac{5}{12}$$

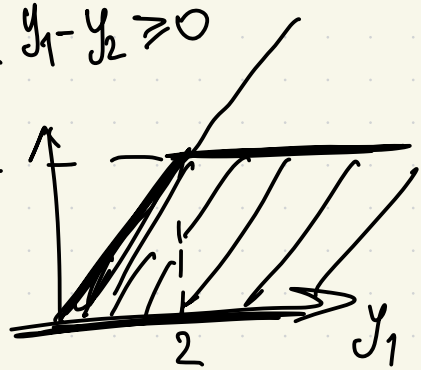
$$D[Y] \approx 0{,}35367$$

③ $X_1 \sim E(2)$, $X_2 \sim U[0;2]$, $S = 4X_1 + X_2$

$$\begin{cases} y_1 = 4x_1 + x_2 \\ y_2 = x_2 \end{cases}, \quad \begin{cases} x_1 = \dfrac{y_1 - y_2}{4} \\ x_2 = y_2 \end{cases} \quad J = \begin{vmatrix} \frac{1}{4} & -\frac{1}{4} \\ 0 & 1 \end{vmatrix} = \frac{1}{4}$$

$$P_{1,2}(y_1, y_2) = \frac{1}{4}\, 2e^{-2\left(\frac{y_1}{4} - \frac{y_2}{4}\right)} \cdot \frac{1}{2} \left| \begin{cases} 0 \le y_2 \le 2 \\ y_1 - y_2 \ge 0 \end{cases} \right.$$

$$P_2(y_1) = \int \frac{1}{4} e^{-\frac{y_1}{2} + \frac{y_2}{2}}\, dy_2 =$$



$$= \begin{cases} \frac{1}{4} e^{-\frac{y_1}{2}} \int_0^{y_1} e^{\frac{y_2}{2}}\, dy, & y \in [0;2] \\ \frac{1}{4} e^{-\frac{y_1}{2}} \int_0^{2} e^{\frac{y_2}{2}}\, dy, & y > 2 \end{cases} =$$

$$= \begin{cases} \frac{1}{4} e^{-\frac{y_1}{2}} \left(2e^{\frac{y_1}{2}} - 1\right), & y \in [0;2] \\ \frac{1}{4} e^{-\frac{y_1}{2}} \left(2e - 1\right), & y > 2 \end{cases}$$

Вариант 3

1. $X \sim E(1)$

② $f(x) = 6x^5 I_{[0;1]}(x)$, $n = 150$

$I \sim Be(0{,}01)$

$X = I \cdot B$

$$E[B] = \int_0^1 6x^5 \cdot x\,dx = \int_0^1 6x^6\,dx = 6\left(\frac{x^7}{7}\right)\Big|_0^1 = \frac{6}{7} = \mu$$

$$E[B^2] = \int_0^1 6x^5 \cdot x^2\,dx = 6\left(\frac{x^8}{8}\right)\Big|_0^1 = \frac{3}{4}$$

$$D[B] = \frac{3}{4} - \left(\frac{6}{7}\right)^2 = \frac{3}{4} - \frac{36}{49} = \frac{3 \cdot 49 - 36 \cdot 4}{4 \cdot 49} =$$

$$= \qquad = \sigma^2$$

$$E[X] = E[I] \cdot E[B] = p \cdot \mu = 0{,}01 \cdot \frac{6}{7} = \frac{6}{700}$$

$$D[X] = \mu^2 p(1-p) + \sigma^2 p =$$

$$= \left(\frac{6}{7}\right)^2 \cdot 0{,}01 \cdot 0{,}99 +$$

$$Y = \sum^n X_i \,, \quad E[Y] = n\,E[X] = \qquad = a$$

$$D[Y] = n\,D[X] = \qquad = b$$

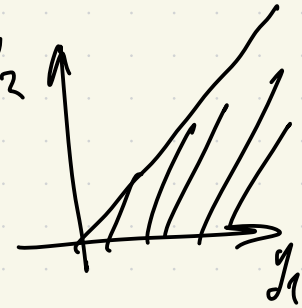
$$P\{y > 1{,}05\,a\} = P\left\{\frac{y-a}{b} > \frac{0{,}05\,a}{b}\right\} =$$

$$= 1 - \Phi\left(\frac{0{,}05\,a}{b}\right)$$

③ $X_1 \sim E(2)$, $X_2 \sim E(5)$, $S = 7X_1 + X_2$

$$\begin{cases} y_1 = 7x_1 + x_2 \\ y_2 = x_2 \end{cases}, \quad \begin{cases} x_1 = \dfrac{y_1 - y_2}{7} \\ x_2 = y_2 \end{cases} \quad J = \frac{1}{7}$$

$$P_{1,2}(y_1, y_2) = \frac{1}{7} f_1\left(\frac{y_1}{7} - \frac{y_2}{7}\right) \cdot f(y_2)$$



$$P_1(y_1) = \int_0^{y_1} \frac{1}{7}\, 2e^{-2\left(\frac{y_1}{7} - \frac{y_2}{7}\right)} \cdot 5e^{-5y_2}\, dy_2 =$$

$$\begin{cases} y_1 - y_2 \geq 0 \\ y_2 \geq 0 \end{cases} \quad = \frac{10}{7} e^{-\frac{2}{7}y_1} \int_0^{y_1} e^{\frac{2}{7}y_2 - 5y_2}\, dy_2 =$$

$$= \frac{10}{7} e^{-\frac{2}{7}y_1} \int_0^{y_1} e^{-\frac{33}{7}y_2}\, dy_2 = \frac{10}{7} e^{-\frac{2}{7}y_1} \left(-\frac{7}{33} e^{-\frac{33}{7}y_2}\right)\Bigg|_0^{y_1} =$$

$$= \frac{10}{7} e^{-\frac{2}{7}y_1} \left(-\frac{7}{33} e^{-\frac{33}{7}y_1} + \frac{7}{33}\right) = \boxed{\frac{10}{33} e^{-\frac{2}{7}y_2} \left(1 - e^{-\frac{33}{7}y_1}\right)}$$